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Inventor(s)	Suzuki; Katsutoshi

Lens holder driving apparatus

Abstract

A lens holder driving apparatus includes a fixed member; a lens holder configured to hold a lens body; a shaft member provided on the fixed member and extending along an optical axis direction so as to guide the lens holder; a piezoelectric driver including a piezoelectric element and configured to move the lens holder along the optical axis direction by a movement of the piezoelectric element; a receiving member provided on the lens holder; and a preload member provided on the fixed member. The piezoelectric driver is provided on the fixed member. The receiving member is disposed facing the piezoelectric driver so as to contact the piezoelectric driver, and receives the movement of the piezoelectric element. The piezoelectric driver includes a contact member and is configured to be preloaded toward the receiving member by the preload member such that the receiving member contacts the contact member.

Inventors:	Suzuki; Katsutoshi (Fukushima, JP)
Applicant:	ALPS ALPINE CO., LTD. (Tokyo, JP)
Family ID:	85386580
Assignee:	ALPS ALPINE CO., LTD. (Tokyo, JP)
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Primary Examiner: Pham; Thomas K

Assistant Examiner: Edenfield; Kuei-Jen L

Attorney, Agent or Firm: IPUSA, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims priority to Japanese Patent Application No. 2021-147092, filed on Sep. 9, 2021, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The disclosure herein relates to a lens holder driving apparatus.

2. Description of the Related Art

(3) An apparatus configured to move a lens holder by using a piezoelectric element is known (see Patent Document 1).

(4) However, in the above-described apparatus, the piezoelectric element is attached to the lens holder that serves as a movable member. Therefore, flexible wiring is required to supply power to the piezoelectric element, thus making it difficult to easily assemble the apparatus.

RELATED-ART DOCUMENTS

Patent Documents

(5) Patent Document 1: Japanese Laid-open Patent Publication No. 2006-098580

SUMMARY OF THE INVENTION

(6) It is desirable to provide a lens holder driving apparatus that can be easily assembled.

(7) According to an embodiment of the present invention, a lens holder driving apparatus includes a fixed member; a lens holder configured to hold a lens body; a shaft member that is provided on the fixed member and extends along an optical axis direction so as to guide the lens holder; a piezoelectric driver that includes a piezoelectric element and is configured to move the lens holder along the optical axis direction by a movement of the piezoelectric element, a receiving member that is provided on the lens holder; and a preload member that is provided on the fixed member. The piezoelectric element extends in a first direction that intersects the optical axis direction. The piezoelectric driver is provided on the fixed member. The receiving member faces the piezoelectric driver so as to contact the piezoelectric driver, extends in a second direction that intersects the first direction in which the piezoelectric element extends, and receives the movement of the piezoelectric element. The piezoelectric driver includes a contact member on a surface on a lens holder side of the piezoelectric element, and is configured to be preloaded toward the receiving member by the preload member such that the receiving member contacts the contact member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Other objects and further features of the present invention will be apparent from the following detailed description when read in conjunction with the accompanying drawings, in which:

(2) FIG. 1A is a perspective view of a lens holder driving apparatus;

(3) FIG. 1B is an exploded perspective view of the lens holder driving apparatus;

(4) FIG. 2 is a schematic view of a camera module;

(5) FIG. 3 is an exploded perspective view of a lower member;

(6) FIG. 4A is a top view of a lens holder supported by a shaft member;

(7) FIG. 4B is an exploded perspective view of the lens holder;

(8) FIG. 5A is a perspective view of a piezoelectric driver pressed against a receiving member by a preload member;

(9) FIG. 5B is an exploded perspective view of the piezoelectric driver pressed against the receiving member by the preload member;

(10) FIG. 6 is a perspective view of the preload member attached to a base member;

(11) FIG. 7 is a rear view of a structure housed in a housing;

(12) FIG. 8A is a top perspective view of a magnetic field generating member before a magnetic

member is attached to a magnet;

(13) FIG. **8B** is a bottom perspective view of the magnetic field generating member after the magnetic member is attached to the magnet;

(14) FIG. **8C** is a right side view of a magnetic sensor and the magnetic field generating member;

(15) FIG. **8D** is a bottom view of the magnetic sensor and the magnetic field generating member;

(16) FIG. **9** is a perspective view of the lens holder, the shaft member, the receiving member, a lens body, and the magnetic field generating member; and

(17) FIG. **10** is a bottom perspective view of the lens holder.

DESCRIPTION OF THE EMBODIMENTS

(18) In the following, a lens holder driving apparatus **101** according to an embodiment of the present invention will be described with reference to the accompanying drawings. FIG. **1A** is a perspective view of the lens holder driving apparatus **101**. FIG. **1B** is an exploded perspective view of the lens holder driving apparatus **101**. FIG. **2** is a schematic view of a camera module CM in a portable device with a camera, in which the lens holder driving apparatus **101** is included.

(19) In the illustrated example, X1 represents one direction of the X axis of the three-dimensional Cartesian coordinate system, and X2 represents the other direction of the X axis. Y1 represents one direction of the Y axis of the three-dimensional Cartesian coordinate system, and Y2 represents the other direction of the Y axis. Similarly, Z1 represents one direction of the Z axis of the three-dimensional Cartesian coordinate system, and Z2 represents the other direction of the Z axis.

Further, the X1 side of the lens holder driving apparatus **101** corresponds to the front side (subject side) of the lens holder driving apparatus **101**, the X2 side of the lens holder driving apparatus **101** corresponds to the rear side (imaging element side) of the lens holder driving apparatus **101**.

Further, the Y1 side of the lens holder driving apparatus **101** corresponds to the left side of the lens holder driving apparatus **101**, and the Y2 side of the lens holder driving apparatus **101** corresponds to the right side of the lens holder driving apparatus **101**. Further, the Z1 side of the lens holder driving apparatus **101** corresponds to the upper side of the lens holder driving apparatus **101**, and the Z2 side of the lens holder driving apparatus **101** corresponds to the lower side of the lens holder driving apparatus **101**. The same applies to the other drawings.

(20) The lens holder driving apparatus **101** is configured to move a lens body LS along an optical axis OA of the lens body LS.

(21) The lens body LS is an example of an optical member, and includes one or more lenses. Typically, the lens body LS is a cylindrical lens barrel with at least one lens, and the central axis of the lens body LS is configured to extend along the optical axis OA. In the illustrated example, the lens body LS includes a first lens body LS1 constituting a zoom lens and a second lens body LS2 constituting a focus lens. In the illustrated example, the lens body LS includes a first lens body LS1 constituting a zoom lens and a second lens body LS2 constituting a focus lens.

(22) The lens holder driving apparatus **101** is configured to move the lens body LS along an optical axis direction by a piezoelectric driver PD (see FIG. **3**) housed in a housing HS. The optical axis direction includes a direction of the optical axis OA of the lens body LS and a direction parallel to the optical axis OA. Specifically, as indicated by a double arrow AR1 in FIG. **1B** and FIG. **2**, the lens holder driving apparatus **101** can move the first lens body LS1 along the optical axis direction, and as indicated by a double arrow AR2, the lens holder driving apparatus **101** can move the second lens body LS2 along the optical axis direction. That is, the lens holder driving apparatus **101** can separately move the first lens body LS1 and the second lens body LS2 along the optical axis direction.

(23) The housing HS is a part of a fixed member FB, and includes a cover member **1** and a base member **2**.

(24) As illustrated in FIG. **2**, the lens holder driving apparatus **101** is used in the camera module CM such as a periscope-type camera module. In the example illustrated in FIG. **2**, the camera module CM mainly includes a mirror MR, the lens body LS, the lens holder driving apparatus **101**,

an imaging element IS, and the like. The mirror MR may be a prism. In the example illustrated in FIG. 2, the mirror MR is configured to provide a flat reflective surface.

(25) Typically, as illustrated in FIG. 2, the lens holder driving apparatus **101** is disposed at a position farther from a subject than the mirror MR and is configured to allow light LT, coming from the subject and reflected by the mirror MR, to reach the imaging element IS through the lens body LS.

(26) Next, referring to FIG. 1B and FIG. 3, an outline of the lens holder driving apparatus **101** will be described. FIG. 1B is an exploded perspective view of the lens holder driving apparatus **101** in which the cover member **1** is separated from a lower member LM. FIG. 3 is an exploded perspective view of the lower member LM. In FIG. 3, a movable member MB is separated from the fixed member FB. The movable member MB is configured to be moved in the optical axis direction by the piezoelectric driver PD.

(27) As illustrated in FIG. 1B, the lens holder driving apparatus **101** includes the cover member **1** and the lower member LM, which are part of the fixed member FB (housing HS). The cover member **1** is configured to cover the lower member LM. In the present embodiment, the cover member **1** is formed of a synthetic resin material.

(28) The base member **2** is a member constituting a part of the housing HS. In present embodiment, the base member **2** is formed of a synthetic resin material, similar to the cover member **1**.

(29) Specifically, as illustrated in FIG. 3, the base member **2** has a bottomed box-like outer shape that defines a housing portion 2S. The base member **2** has an outer wall portion 2A having a substantially rectangular cylindrical shape and a bottom surface portion 2B having a substantially rectangular flat shape. The bottom surface portion 2B continues with the lower end (Z2 side end) of the outer wall portion 2A. The outer wall portion 2A includes a first side plate portion 2A1, a second side plate portion 2A2, a third side plate portion 2A3, and a fourth side plate portion 2A4. The first side plate portion 2A1 and the third side plate portion 2A3 face each other, and the second side plate portion 2A2 and the fourth side plate portion 2A4 face each other. Further, the second side plate portion 2A2 and the fourth side plate portion 2A4 are perpendicular to the first side plate portion 2A1 and the third side plate portion 2A3. In other words, the first side plate portion 2A1 and the third side plate portion 2A3 are perpendicular to the second side plate portion 2A2 and the fourth side plate portion 2A4. The first side plate portion 2A1 has a circular through hole OP1 for receiving the light LT coming from the subjected and reflected by the mirror MR. Similarly, the third side plate portion 2A3 has a circular through hole OP2 for allowing the light LT to reach the imaging element IS. The cover member **1** is bonded to the base member **2** with an adhesive or the like, and constitutes the housing HS together with the base member **2**.

(30) As illustrated in FIG. 3, a notch 2T, through which a terminal portion 4T of a substrate **4** passes, is formed in the fourth side plate portion 2A4 of the base member **2**. As illustrated in FIG. 1A and FIG. 1B, the base member **2** includes a plug member 2P for closing the notch 2T. In the illustrated example, the plug member 2P is fitted into the notch 2T and fixed with an adhesive. Note that the plug member 2P is not depicted in FIG. 3 for clarity.

(31) As illustrated in FIG. 3, the lower member LM includes a lens holder **3**, which is an example of the movable member MB. The lower member LM further includes the base member **2**, the substrate **4**, a shaft member **5**, and the piezoelectric driver PD which are an example of the fixed member FB.

(32) The lens holder **3** is configured to hold the lens body LS. In the illustrated example, the lens holder **3** is formed by performing injection molding on a synthetic resin such as a liquid crystal polymer (LCP). Further, the lens holder **3** includes a first lens holder 3A configured to hold the first lens body LS1 and a second lens holder 3B configured to hold the second lens body LS2. Note that the first lens holder 3A and the second lens holder 3B have the same shape and the same size. Therefore, the first lens holder 3A and the second lens holder 3B can be formed of the same material, thus reducing the cost.

(33) Next, the lens holder 3 will be described in detail with reference to FIG. 4A and FIG. 4B. FIG. 4A is a top view of the lens holder 3 supported by the shaft member 5. FIG. 4B is an exploded perspective view of the second lens holder 3B. The description with reference to FIG. 4B relates to the second lens holder 3B, but also applies analogously to the first lens holder 3A. In FIG. 4A, the first lens holder 3A is indicated by a coarse dot pattern, and the second lens holder 3B is indicated by a fine dot pattern for clarity. In FIG. 4B, the second lens holder 3B is indicated by a fine dot pattern for clarity.

(34) The shaft member 5 includes a first shaft member 5A having an axis (axis 5AX) parallel to the optical axis OA and a second shaft member 5B having an axis (axis 5BX) parallel to the optical axis OA. Therefore, the first shaft member 5A and the second shaft member 5B extend in the optical axis direction while being parallel to each other. In the illustrated example, the shaft member 5 is configured such that one end of the shaft member 5 is fitted into a groove formed in the first side plate portion 2A1 of the base member 2 and the other end of the shaft member 5 is fitted into a groove formed in the third side plate portion 2A3 of the base member 2. However, the shaft member 5 may be configured such that one end of the shaft member 5 is inserted into a circular through hole (not illustrated) formed in the first side plate portion 2A1 of the base member 2 and the other end of the shaft member 5 is inserted into a circular through hole (not illustrated) formed in the third side plate portion 2A3 of the base member 2. The shaft member 5 may be fixed to the base member 2 with an adhesive.

(35) As illustrated in FIG. 4A and FIG. 4B, the lens holder 3 includes a holding portion 31 that holds the lens body LS, and a shaft-receiving portion 32 that receives the shaft member 5. The shaft-receiving portion 32 includes a left shaft-receiving portion 32L that receives the first shaft member 5A, and a right shaft-receiving portion 32R that receives the second shaft member 5B. The shaft-receiving portion 32 functions as a rotation stopper that stops the rotation of the lens holder 3 around the shaft member 5.

(36) Specifically, the first lens holder 3A includes a first holding portion 31A that holds the first lens body LS1, and a first shaft-receiving portion 32A that receives the shaft member 5. The first shaft-receiving portion 32A includes the left shaft-receiving portion 32L (first left shaft-receiving portion 32AL) that receives the first shaft member 5A, and the right shaft-receiving portion 32R (a first right shaft-receiving portion 32AR) that receives the second shaft member 5B.

(37) Similarly, the second lens holder 3B includes a second holding portion 31B that holds the second lens body LS2, and a second shaft-receiving portion 32B that receives the shaft member 5. The second shaft-receiving portion 32B includes the left shaft-receiving portion 32L (second left shaft-receiving portion 32BL) that receives the first shaft member 5A, and the right shaft-receiving portion 32R (a second right shaft-receiving portion 32BR) that receives the second shaft member 5B.

(38) More specifically, as illustrated in FIG. 4B, the second right shaft-receiving portion 32BR has a circular through hole TH capable of receiving the second shaft member 5B. The same applies to the first left shaft-receiving portion 32AL.

(39) As illustrated in FIG. 10, the second left shaft-receiving portion 32BL has a notch CT that opens to the left (in the Y1 direction) and that is capable of receiving the first shaft member 5A. The same applies to the first right shaft-receiving portion 32AR.

(40) A receiving member 7 is fixed to one of the two shaft-receiving portions 32 of each of the first lens holder 3A and the second lens holder 3B. The receiving member 7 is a member disposed in contact with a contact member 9 (see FIG. 3) of piezoelectric driver PD. The receiving member 7 includes a first receiving member 7A having an axis 7AX parallel to the optical axis OA, and a second receiving member 7B having an axis 7BX parallel to the optical axis OA.

(41) In the illustrated example, the receiving member 7 is formed of a material different from that of the lens holder 3, and has a rod shape (substantially cylindrical shape). Specifically, the lens holder 3 is formed of a liquid crystal polymer (LCP), and the receiving member 7 is formed of a

metal. However, the receiving member 7 may be a part of the lens holder 3. In this case, the receiving member 7 may be formed of the same material as that of the lens holder 3, and may be integrated with the lens holder 3. Further, the receiving member 7 may have any other shape such as an elliptical column shape or a rectangular column shape. Alternatively, the receiving member 7 may have a plate shape.

(42) In the example illustrated in FIG. 4A and FIG. 4B, the first receiving member 7A is fixed to the first left shaft-receiving portion 32AL of the first lens holder 3A, and the second receiving member 7B is fixed to the second right shaft-receiving portion 32BR of the second lens holder 3B. That is, in the example illustrated in FIG. 4A, the first lens holder 3A and the second lens holder 3B are disposed to have twofold rotational symmetry with respect to a center line CL (a line parallel to the Z axis) of the lens holder driving apparatus 101.

(43) Further, the shaft-receiving portion 32, to which the receiving member 7 is fixed, is formed such that the length of the shaft-receiving portion 32 in the optical axis direction is greater than that of the holding portion 31. In the example illustrated in FIG. 4A, the first left shaft-receiving portion 32AL, to which the first receiving member 7A is fixed, is formed such that the width W1, which is a dimension in the optical axis direction, of the first left shaft-receiving portion 32AL is greater than the width W2 of the first holding portion 31A and greater than the width W3 of the first receiving member 7A.

(44) Conversely, the first right shaft-receiving portion 32AR is formed such that its width W4 is smaller than the width W2 of the first holding portion 31A. The first right shaft-receiving portion 32AR is formed at a front end portion (an end portion on the X1 side) of the first holding portion 31A, such that the first right shaft-receiving portion 32AR and the second right shaft-receiving portion 32BR do not make contact with each other when the first lens holder 3A and the second lens holder 3B approach each other.

(45) Similarly, the right shaft-receiving portion 32BR, to which the second receiving member 7B is fixed, is formed such that its width W5 is greater than the width W6 of the second holding portion 31B and greater than the width W7 of the second receiving member 7B.

(46) Conversely, the second left shaft-receiving portion 32BL is formed such that its width W8 is smaller than the width W6 of the second holding portion 31B. The second left shaft-receiving portion 32BL is formed at a rear end portion (an end portion on the X2 side) of the second holding portion 31B, such that the first left shaft-receiving portion 32AL and the second left shaft-receiving portion 32BL do make contact with each other when the first lens holder 3A and the second lens holder 3B approach each other.

(47) With the above-described configuration, the first lens holder 3A and the second lens holder 3B may be disposed, such that at least a part of the first left shaft-receiving portion 32AL and a part of the second right shaft-receiving portion 32BR overlap in the y-axis direction in a state in which the first holding portion 31A and the second holding portion 31B are in proximity to each other. In the state illustrated in FIG. 4A, the first left shaft-receiving portion 32AL of the first lens holder 3A and the second right shaft-receiving portion 32BR of the second lens holder 3B are partially at the same position in the optical axis direction. Accordingly, the first receiving member 7A and the second receiving member 7B overlap over the width W9 in the y-axis direction.

(48) Accordingly, one effect of the above-described configuration is that the first lens body LS1 and the second lens body LS2 can approach each other to the extent that the first lens body LS1 and the second lens body LS2 are nearly in contact with each other, while the width W1 of the first left shaft-receiving portion 32AL is set to be greater than the width W2 of the first holding portion 31A and the width W5 of the second right shaft-receiving portion 32BR is set to be greater than the width W6 of the second holding portion 31B.

(49) In the illustrated example, the first lens holder 3A and the second lens holder 3B are configured to have the same size and the same shape. Therefore, the width W1 is equal to the width W5, the width W2 is equal to the width W6, the width W3 is equal to the width W7, and the width

W4 is equal to the width W8. However, the width W1 may be different from the width W5, the width W2 may be different from the width W6, the width W3 may be different from the width W7, and the width W4 may be different from the width W8.

(50) Next, the piezoelectric driver PD will be described with reference to FIG. 5A, FIG. 5B, and FIG. 6. FIG. 5A is a perspective view of the piezoelectric driver PD pressed against the receiving member 7 by a preload member 13. FIG. 5B is an exploded perspective view of the piezoelectric driver PD pressed against the receiving member 7 by the preload member 13. FIG. 6 is a perspective view of the preload member 13 attached to the base member 2. In FIG. 6, the preload member 13 is indicated by a dot pattern for clarity.

(51) The piezoelectric driver PD is configured to move the lens holder 3 along the optical axis direction. In the present embodiment, the piezoelectric driver PD is an example of a frictional driving unit that uses a driving system as described in U.S. Pat. No. 7,786,648. The piezoelectric driver PD includes a piezoelectric element 8, the contact member 9, and a circuit board 10.

(52) The piezoelectric driver PD is configured to be preloaded inward (toward the optical axis OA) by the preload member 13, which is fixed to the base member 2, and to be pressed against the receiving member 7. In the illustrated example, the preload member 13 is formed of a metal plate, and is configured so as to contact the outer surface (on the side far from the optical axis OA) of the piezoelectric element 8 through the circuit board 10, at a position (an inner edge portion BE) corresponding to each of two nodes ND (see FIG. 5B) of flexural vibration (a circular motion as will be described later) of the piezoelectric element 8. The preload member 13 is bonded to the piezoelectric driver PD with an adhesive, for example.

(53) As illustrated in FIG. 3, the piezoelectric driver PD includes a first piezoelectric driver PD1 configured to move the first lens holder 3A along the optical axis direction, and a second piezoelectric driver PD2 configured to move the second lens holder 3B along the optical axis direction. The first piezoelectric driver PD1 includes a first piezoelectric element 8A, a first contact member 9A, and a first circuit board 10A. The second piezoelectric driver PD2 includes a second piezoelectric element 8B, a second contact member 9B, and a second circuit board 10B.

(54) The preload member 13 includes a first preload member 13A disposed to press the first piezoelectric driver PD1 against the first receiving member 7A, and a second preload member 13B disposed to press the second piezoelectric driver PD2 against the second receiving member 7B.

(55) In FIG. 5A and FIG. 5B, the first piezoelectric driver PD1 pressed against the first receiving member 7A by the first preload member 13A is depicted. However, the following description with reference to FIG. 5A and FIG. 5B also applies analogously to the second piezoelectric driver PD2 pressed against the second receiving member 7B by the second preload member 13B. This is because the first piezoelectric driver PD1 and the second piezoelectric driver PD2 have the same configuration.

(56) In the illustrated example, the first piezoelectric element 8A extends in the Z axis direction perpendicular to the optical axis direction (the X-axis direction) (that is, the direction perpendicular to the optical axis OA), and is configured to generate a flexural vibration (circular motion) having two nodes ND. Specifically, the first piezoelectric element 8A has a two-layer structure including a first layer for generating a first flexural vibration in the XZ plane and a second layer for generating a second flexural vibration in the YZ plane. The first piezoelectric driver PD1 applies a voltage to the a piezoelectric element constituting the first layer and a voltage to a piezoelectric element constituting the second layer individually at appropriate timings, so that a locus drawn by the center point of the first piezoelectric element 8A becomes a circular orbit around a rotation axis 8AX. That is, the first piezoelectric element 8A can generate a movement (circular motion) such that the center point of the first piezoelectric element 8A draws a circle. In the example illustrated in FIG. 5B, the rotation axis 8AX is parallel to the Z axis. By applying the voltages at appropriate timings, the first piezoelectric driver PD1 can switch the moving direction (rotational direction) of the center point, following the circular orbit, between the clockwise direction and the counterclockwise

direction as viewed from the Z1 side. Note that the circle (circular orbit) drawn by the center point of the first piezoelectric element **8A** is not necessarily a perfect circle, and may be an approximate circle.

(57) In FIG. 5B, a dotted arrow drawn around the first piezoelectric element **8A** indicates the flexural vibration (circular motion) of the first piezoelectric element **8A** (that is, the first piezoelectric element **8A** rotates in the clockwise direction around the rotation axis **8AX** while deflecting). In this case, the first receiving member **7A** (first lens holder **3A**) that contacts the first contact member **9A** of the first piezoelectric driver PD1 including the first piezoelectric element **8A** moves forward (in the X1 direction). Although not illustrated, the first piezoelectric element **8A** can also rotate in the counterclockwise direction around the rotation axis **8AX** while deflecting. In this case, the first receiving member **7A** (the first lens holder **3A**) that contacts the first contact member **9A** moves rearward (in the X2 direction).

(58) That is, the first lens holder **3A** (first receiving member **7A**) moves forward (in the X1 direction) when the rotational direction of the center point of the first piezoelectric element **8A** is clockwise, and moves rearward (in the X2 direction) when the rotational direction of the center point of the first piezoelectric element **8A** is counterclockwise. The center point of the first piezoelectric element **8A** is a point at which the amplitude of the first flexural vibration is maximum (a point corresponding to an antinode of the first flexural vibration) and is also a point at which the amplitude of the second flexural vibration is maximum (a point corresponding to an antinode of the second flexural vibration).

(59) The first contact member **9A** is attached to the first piezoelectric element **8A**, and is configured to contact the first receiving member **7A**. In the illustrated example, the first contact member **9A** is bonded to the surface on the inner side (on the Y2 side that faces the optical axis OA) of the first piezoelectric element **8A** with an adhesive so as to cover the entire surface on the inner side of the first piezoelectric element **8A**. The first contact member **9A** is formed of a metal such as stainless steel. The first contact member **9A** has an appropriate thickness such that a flexural vibration (circular motion) can be performed in response to the flexural vibration (circular motion) of the first piezoelectric element **8A**. In the illustrated example, the first contact member **9A** is a friction plate formed of stainless steel. The first contact member **9A** extends in the Z-axis direction that is the extending direction of the first piezoelectric element **8A**. The first contact member **9A** is configured such that a center portion in the extending direction of the first contact member **9A** contacts the first receiving member **7A** attached to the first lens holder **3A**.

Specifically, the first contact member **9A** is configured to contact the first receiving member **7A** at a position where the amplitude of the flexural vibration (circular motion) is maximum (a position corresponding to an antinode of the flexural vibration). Further, the surface on a side in contact with the first receiving member **7A** (that is, the surface on the Y2 side) of the first contact member **9A** has a curved shape (an arc shape). The first receiving member **7A** is formed of a metal such as stainless steel. In the illustrated example, the first receiving member **7A** is a rod member formed of stainless steel, having a cylindrical shape, and extending in the optical axis direction. In order to prevent abrasion of the first lens holder **3A** made of a synthetic resin due to contact with the first contact member **9A** made of a metal, the first contact member **9A** made of a metal contacts the first receiving member **7A** made of a metal. Note that the length of the first contact member **9A** in the Z-axis direction may be different from the length of the first piezoelectric element **8A**, as long as the first contact member **9A** contacts the first receiving member **7A**.

(60) The first circuit board **10A** is a board that includes a conductive pattern, and is configured to electrically connect an external power source to the first piezoelectric element **8A**. In the illustrated example, the first circuit board **10A** is a flexible printed circuit board having flexibility. One end of the first circuit board **10A** is connected to the substrate **4**, and the first circuit board **10A** is configured to apply a voltage to the first piezoelectric element **8A**. The first piezoelectric element **8A** is bonded to the surface on the inner side (on the Y2 side that faces the optical axis OA) of the

first circuit board **10A** with an anisotropic conductive adhesive. The first piezoelectric element **8A** may be bonded to the surface on the inner side of the first circuit substrate **10A** with an anisotropic conductive adhesive film.

(61) In the illustrated example, the preload member **13** is a leaf spring member. Specifically, as illustrated in FIG. 5B, the first preload member **13A** includes a fixed portion **13F** that is fixed to the base member **2**, a support portion **13S** that supports the first piezoelectric driver **PD1**, and an elastically deformable portion **13E** that is elastically deformable and provided between the fixed portion **13F** and the support portion **13S**. Further, in the preload member **13**, the fixed portion **13F** is fixed to the base member **2** such that the support portion **13S** and the elastically deformable portion **13E** do not contact the base member **2**. The following description with reference to FIG. 5B relates to the first preload member **13A**, but also applies analogously to the second preload member **13B**. This is because the second preload member **13B** has the same shape and the same size as those of the first preload member **13A**.

(62) Specifically, the fixed portion **13F** includes a front fixed portion **13FF** and a rear fixed portion **13FB**. The support portion **13S** includes an upper support portion **13SU** and a lower support portion **13SD**. The elastically deformable portion **13E** includes a front elastically deformable portion **13EF** provided between the front fixed portion **13FF** and the upper support portion **13SU**, and a rear elastically deformable portion **13EB** provided between the rear fixed portion **13FB** and the upper support portion **13SU**. The front fixed portion **13FF** and the rear fixed portion **13FB** have the same shape and the same size. The upper support portion **13SU** and the lower support portion **13SD** have the same shape and the same size. The front elastically deformable portion **13EF** and the rear elastically deformable portion **13EB** have the same shape and the same size. That is, the first preload member **13A** is configured to be symmetric with respect to a symmetry plane parallel to the YZ plane (plane that divides the first preload member **13A** into two front and rear portions). The first preload member **13A** is also configured to be symmetric with respect to another symmetry plane parallel to the XY plane (plane that divides the first preload member **13A** into two upper and lower portions). The same applies to the second preload member **13B**.

(63) In the illustrated example, the front elastically deformable portion **13EF** is connected to the front fixed portion **13FF** through a pair of upper and lower elastic arm portions. However, the front elastically deformable portion **13EF** may be connected to the front fixed portion **13FF** through one elastic arm portion or three or more elastic arm portions. Further, although the elastic arm portions are configured to extend linearly along the optical axis direction, the elastic arm portions may have any shape such as a shape including a curved portion. The same applies to the rear elastically deformable portion **13EB**. In the illustrated example, the front fixed portion **13FF** is configured such that the ends of the front fixed portion **13FF** are connected to the respective upper and lower elastic arm portions of the front elastically deformable portion **13EF**. However, the ends of the front fixed portion **13FF** are not necessarily connected to the respective upper and lower elastic arm portions. That is, the front fixed portion **13FF** may be configured by two or more separate portions. The same applies to the rear fixed portion **13FB**.

(64) The support portion **13S** includes a base portion **13K** and a bent portion **13N**. The base portion **13K** is connected to the elastically deformable portion **13E**, and the bent portion **13N** is bent in an L-shape from the base portion **13K** and protrudes toward the lens holder **3** (toward the Y2 side). Further, a recess **RS** is formed in the tip of the bent portion **13N**. The recess **RS** is a recess that opens toward the lens holder **3** (toward the Y2 side). Specifically, the base portion **13K** includes an upper base portion **13KU**, which is a part of the upper support portion **13SU**, and a lower base portion **13KD**, which is a part of the lower support portion **13SD**. The bent portion **13N** includes an upper bent portion **13NU**, which is a part of the upper support portion **13SU**, and a lower bent portion **13ND**, which is a part of the lower support portion **13SD**. Recesses **RS** are formed in the tips of the upper bent portion **13NU** and the lower bent portion **13ND**, and have the same shape and the same size. As illustrated in FIG. 5A, a part of the first piezoelectric driver **PD1** is disposed in a

recess RS, and is fixed to the bent portion **13N** with an adhesive while contacting the inner edge portion BE of the recess RS.

(65) More specifically, as illustrated in FIG. 5B, the recess RS has a front edge portion and a rear edge portion that face each other across the inner edge portion BE. Further, as illustrated in FIG. 5A, the first piezoelectric driver PD1 is disposed between the front edge portion and the rear edge portion.

(66) A position at the first piezoelectric driver PD1 contacts the inner edge portion BE of the recess RS corresponds to the position of each of the nodes ND of the first piezoelectric element **8A** that generates the flexural vibration. The position of each of the nodes ND includes the position of a first node ND1 and the position of a second node ND2. In FIG. 5B, the position of each of the nodes ND is indicated by a cross pattern for clarity.

(67) The position at which the first piezoelectric driver PD1 contacts the inner edge portion BE of the recess RS (that is, the position of each of the nodes ND) corresponds to a position a predetermined distance from an end portion of the first piezoelectric driver PD1. The predetermined distance is, for example, approximately one quarter of the total length of the piezoelectric driver PD.

(68) The first piezoelectric driver PD1 is fixed to the bent portion **13N** with an adhesive. Specifically, the first piezoelectric driver PD1 (first circuit board **10A**) is fixed to the inner edge portion BE of the recess RS of the bent portion **13N** with an adhesive. Further, the first circuit board **10A** of the first piezoelectric driver PD1 is fixed to a contact portion AF of the front elastically deformable portion **13EF** and to a contact portion AB of the rear elastically deformable portion **13EB** of the first preload member **13A** with an adhesive. In the illustrated example, an ultraviolet curable adhesive is used; however, any other adhesive such as a moisture curable adhesive or a thermosetting adhesive may be used.

(69) As illustrated in FIG. 5B, the elastically deformable portion **13E** extends away from the base portion **13K**. Specifically, the front elastically deformable portion **13EF** extends forward (in the X1 direction) from the upper base portion **13KU** and the lower base portion **13KD**, and the rear side elastically deformable portion **13EB** extends rearward (in the X2 direction) from the upper base portion **13KU** and the lower base portion **13KD**. The extending direction of the elastically deformable portion **13E** is parallel to the optical axis direction.

(70) The fixed portion **13F** is provided on an extension line of the elastically deformable portion **13E**. As illustrated in FIG. 6, the fixed portion **13F** is held by a holding portion **2C** provided on the base member **2**. Specifically, the holding portion **2C** has a groove **2G** into which the fixed portion **13F** is fitted. In the illustrated example, the fixed portion **13F** of the preload member **13** is held by the holding portion **2C** by being fitted into the groove **2G** of the holding portion **2C** from above. Note that the fixed portion **13F** may be held by the holding portion **2C** with an adhesive, or the fixed portion **13F** held by the holding portion **2C** may be reinforced with an adhesive.

(71) Next, a position detection mechanism DT will be described with reference to FIG. 7. FIG. 7 is a rear view of a structure housed in the housing HS that includes the cover member **1** and the base member **2**. The structure housed in the housing HS includes the position detection mechanism DT, the lens body LS, the piezoelectric driver PD, the lens holder **3**, the substrate **4**, and the preload member **13**. In FIG. 7, for clarity, the substrate **4** is not depicted, and the first lens holder **3A** is indicated by a coarse dot pattern and the second lens holder **3B** is indicated by a fine dot pattern.

(72) The position detection mechanism DT is a mechanism for detecting the position of the lens holder **3**. The position detection mechanism DT includes a magnetic sensor **6** and a magnetic field generating member MG. The magnetic field generating member MG includes a magnet **11** and a magnetic member **12**.

(73) The magnetic sensor **6** is configured to detect a magnetic field generated by the magnetic field generating member MG. In the illustrated example, the magnetic sensor **6** is configured by a giant magnetoresistance (GMR) element. The magnetic sensor **6** is configured to measure a voltage

value, which changes according to the magnitude of a magnetic field generated by the magnetic field generating member MG and is received by the magnetic sensor **6**, and to detect the position of the lens holder **3** to which the magnetic field generating member MG is attached. Further, the magnetic sensor **6** is configured to output a larger voltage value as an N-pole portion approaches and output a smaller voltage value as an S-pole portion approaches. However, the magnetic sensor **6** may be configured to output a smaller voltage value as an N-pole portion approaches and output a larger voltage value as an S-pole portion approaches. The magnetic sensor **6** may be configured to use any other magnetoresistive element, such as a semiconductor magnetoresistive (SMR) element, an anisotropic magnetoresistive (AMR) element, or a tunnel magnetoresistive (TMR) element, to detect the position of the lens holder **3**, or may be configured to use a Hall element to detect the position of the lens holder **3**.

(74) In the illustrated example, the position detection mechanism DT includes a first position detection mechanism DT1 for detecting the position of the first lens holder **3A**, and a second position detection mechanism DT2 for detecting the position of the second lens holder **3B**. The first position detection mechanism DT1 includes a first magnetic sensor **6A** and a first magnetic field generating member MG1. The second position detection mechanism DT2 includes a second magnetic sensor **6B** and a second magnetic field generating member MG2. The first magnetic field generating member MG1 includes a first magnet **11A** and a first magnetic member **12A**. The second magnetic field generating member MG2 includes a second magnet **11B** and a second magnetic member **12B**. The first magnetic member **12A** includes a first left magnetic member **12AL** and a first right magnetic member **12AR**. The second magnetic member **12B** includes a second left magnetic member **12BL** and a second right magnetic member **12BR**.

(75) The magnetic field generating member MG will be described in detail with reference to FIG. **8A** through FIG. **8D**. FIG. **8A** is a top perspective view of the magnetic field generating member MG before the magnetic member **12** is attached to the magnet **11**. FIG. **8B** is a bottom perspective view of the magnetic field generating member MG after the magnetic member **12** is attached to the magnet **11**. FIG. **8C** and FIG. **8D** illustrate the positional relationship between the magnetic sensor **6** and the magnetic field generating member MG, which changes as the lens holder **3** moves in the optical axis direction. Specifically, FIG. **8C** is a right side view of the magnetic sensor **6** and the magnetic field generating member MG, and FIG. **8D** is a bottom view of the magnetic sensor **6** and the magnetic field generating member MG. FIG. **8A** through FIG. **8D** relate to the second magnetic field generating member MG2 that is attached to the lower surface of the second right shaft-receiving portion **32BR** of the second lens holder **3B**. In the illustrated example, the second magnetic field generating member MG2 is fixed to the lower surface of the second right shaft-receiving portion **32BR** of the second lens holder **3B** with an adhesive. The following description with reference to FIG. **8A** through FIG. **8D** also applies analogously to the first magnetic field generating member MG1 that is attached to the lower surface of the first left shaft-receiving portion **32AL** of the first lens holder **3A**.

(76) As illustrated in FIG. **8A**, the second magnetic field generating member MG2 includes the second magnet **11B**, the second left magnetic member **12BL**, and the second right magnetic member **12BR**.

(77) The second magnet **11B** is a permanent magnet magnetized to two poles in the Y-axis direction, which is a direction perpendicular to the optical axis direction. In the illustrated example, the second magnet **11B** is magnetized such that one end portion (a left end portion) in the direction (Y-axis direction) perpendicular to the optical axis direction becomes an N-pole and the other end portion (a right end portion) becomes an S-pole. In FIG. **8A**, for clarity, the N-pole portion of the second magnet **11B** is indicated by a fine cross pattern, and the S-pole portion of the second magnet **11B** is indicated by a coarse cross pattern.

(78) In FIG. **8B**, the second left magnetic member **12BL** and the second right magnetic member **12BR** are both connected to the second magnet **11B**. In the illustrated example, the second left

magnetic member **12BL** and the second right magnetic member **12BR** are fixed to the second magnet **11B** with an adhesive.

(79) Specifically, the second left magnetic member **12BL** is connected to the N-pole portion, which is the one end portion (left end portion) of the second magnet **11B**, and is magnetized to the N-pole, and the second right magnetic member **12BR** is connected to the S-pole portion, which is the other end portion (right end portion) of the second magnet **11B**, and is magnetized to the S-pole.

Therefore, in FIG. **8B**, similar to the N-pole portion of the second magnet **11B**, the second left magnetic member **12BL** is indicated by a fine cross pattern, and, similar to the S-pole portion of the second magnet **11B**, the second right magnetic member **12BR** is indicated by a rough cross pattern. The same applies to FIG. **8C** and FIG. **8D**. Note that, in the state illustrated in FIG. **8A**, the second left magnetic member **12BL** and the second right magnetic member **12BR** are not indicated by cross patterns because the second left magnetic member **12BL** and the second right magnetic member **12BR** are not connected to the second magnet **11B** and are not magnetized.

(80) In the illustrated Example, as illustrated in FIG. **8A**, the second magnetic member **12B** includes a rod portion **BP** and a protruding portion **PR** having a L-shape. The protruding portion **PR** protrudes downward (in the **Z2** direction) from the rod portion **BP**, and subsequently protrudes inward. Specifically, the second left magnetic member **12BL** includes a left rod portion **BPL** and three protruding portions **PR** (first to third left protruding portions **PR1L** to **PR3L**). The three protruding portions **PR** (first to third left protruding portions **PR1L** to **PR3L**) protrude downward from the left rod portion **BPL**, and subsequently protrude toward the second magnet **11B** (in the **Y2** direction). Further, the second right magnetic member **12BR** includes a right rod portion **BPR** and three protruding portions **PR** (first to third right protruding portions **PR1R** to **PR3R**). The three protruding portions **PR** (first to third right protruding portions **PR1R** to **PR3R**) protrude downward from the right rod portion **BPR**, and subsequently protrude toward the second magnet **11B** (in the **Y1** direction).

(81) In the illustrated Example, the second left magnetic member **12BL** is configured such that the height, which is a dimension in the **Z**-axis direction, of the left rod portion **BPL**, is the same as the height of the second magnet **11B**. However, the height of the left rod portion **BPL** may be greater than the height of the second magnet **11B**. In this case, the second left magnetic member **12BL** may be connected to the second magnet **11B** such that the upper end surface of the left rod portion **BPL** is coplanar with the upper end surface of the second magnet **11B**. Further, each of the three protruding portions **PR** may be configured to protrude toward the second magnet **11B** (in the **Y2** direction) from a portion, of the left rod portion **BPL**, that protrudes downward (in the **Z2** direction) relative to the lower end surface of the second magnet **11B**. The same applies to the second right magnetic member **12BR**. Alternatively, the second left magnetic member **12BL** having such a shape may be connected to the second magnet **11B** such that the upper end surface of the left rod portion **BPL** contacts the lower end surface of the second magnet **11B**. In this case, the width, which is a dimension in the **Y**-axis direction, of the second magnet **11B** may be increased. The same applies to the second right magnetic member **12BR**.

(82) Further, in the illustrated Example, the second left magnetic member **12BL** and the second right magnetic member **12BR** are configured, such that the first to third left protruding portions **PR1L** to **PR3L** and the first to third right protruding portions **PR1R** to **PR3R** are alternately arranged in the extending direction (**X**-axis direction) of the second magnetic field generating member **MG2**.

(83) Specifically, as illustrated in FIG. **8B**, the second left magnetic member **12BL** and the second right magnetic member **12BR** are configured such that the first right protruding portion **PR1R** is disposed on the frontmost side (**X1** side) in the extending direction (**X**-axis direction) of the second magnetic field generating member **MG2**, the first left protruding portion **PR1L** is disposed rearward (at the **X2** side) of the first right protruding portion **PR1R**, the second right protruding portion **PR2R** is disposed rearward (at the **X2** side) of the first left protruding portion **PR1L**, the second left

protruding portion PR2L is disposed rearward (at the X2 side) of the second right protruding portion PR2R, the third right protruding portion PR3R is disposed rearward (at the X2 side) of the second left protruding portion PR2L, and the third left protruding portion PR3L is disposed rearward (at the X2 side) of the third right protruding portion PR3R.

(84) Further, as illustrated in FIG. 8A, the second left magnetic member 12BL and the second right magnetic member 12BR are configured such that the width M1 of the first left protruding portion PR1L, the width M2 of the second left protruding portion PR2L, the width M3 of the third left protruding portion PR3L, the width M4 of the first right protruding portion PR1R, the width M5 of the second right protruding portion PR2R, and the width M6 of the third right protruding portion PR3R in the extending direction (X-axis direction) of the second magnetic field generating member MG2 are the same.

(85) Further, as illustrated in FIG. 8A, the second left magnetic member 12BL and the second right magnetic member 12BR are configured, such that the distance L1 between the first left protruding portion PR1L and the second left protruding portion PR2L, the distance L2 between the second left protruding portion PR2L and the third left protruding portion PR3L, the distance L3 between the first right protruding portion PR1R and the second right protruding portion PR2R, and the distance L4 between the second right protruding portion PR2R and the third right protruding portion PR3R in the extending direction (X-axis direction) of the second magnetic field generating member MG2 are the same.

(86) Further, as illustrated in FIG. 8B, the second left magnetic member 12BL and the second right magnetic member 12BR are configured such that the distance N1 between the first right protruding portion PR1R and the first left protruding portion PR1L, the distance N2 between the first left protruding portion PR1L and the second right protruding portion PR2R, the distance N3 between the second right protruding portion PR2R and the second left protruding portion PR2L, the distance N4 between the second left protruding portion PR2L and the third right protruding portion PR3R, and the distance N5 between the third right protruding portion PR3R and the third left protruding portion PR3L in the extending direction (X-axis direction) of the second magnetic field generating member MG2 are the same.

(87) Further, as illustrated in FIG. 8C, the second left magnetic member 12BL and the second right magnetic member 12BR are configured such that the thicknesses TK of portions, protruding toward the second magnet 11B, of the first to third left protruding portions PR1L to PR3L and of the first to third right protruding portions PR1R to PR3R are the same.

(88) Further, as illustrated in FIG. 8C, the second left magnetic member 12BL and the second right magnetic member 12BR are configured, such that the distances GP between the magnetic sensor 6 (second magnetic sensor 6B), fixed to the substrate 4 (not illustrated), and the lower end surfaces (on the Z2 side) of the first to third left protruding portions PR1L to PR3L and of the first to third right protruding portions PR1R to PR3R are the same.

(89) However, the second left magnetic member 12BL and the second right magnetic member 12BR may have different widths M1 to M6. The same applies to the distances L1 to L4, and the same applies to the distances N1 to N5. The second left magnetic member 12BL and the second right magnetic member 12BR may be configured such that the thicknesses TK of the portions, protruding toward the second magnet 11B, of the first to third left protruding portions PR1L to PR3L and of the first to third right protruding portions PR1R to PR3R are different. The same applies to the distances GP.

(90) The middle part of each of FIG. 8C and FIG. 8D illustrates the positional relationship between the second magnetic field generating member MG2 and the second magnetic sensor 6B when the second lens holder 3B is located at a reference position illustrated in FIG. 1B. Each of the upper part of FIG. 8C and the left part of FIG. 8D illustrates the positional relationship between the second magnetic field generating member MG2 and the second magnetic sensor 6B when the second lens holder 3B is moved rearward (in the X2 direction) from the reference position. Each of

the lower part of FIG. 8C and the right part of FIG. 8D illustrates the positional relationship between the second magnetic field generating member MG2 and the second magnetic sensor 6B when the second lens holder 3B is moved forward (in the X1 direction) from the reference position. (91) The second magnetic sensor 6B is provided on the substrate 4 fixed to the base member 2 so as to detect a magnetic field generated by each of the first to third left protruding portions PR1L to PR3L and of the first to third right protruding portions PR1R to PR3R.

(92) When the second lens holder 3B is located at the reference position, the second magnetic sensor 6B partially faces the second right protruding portion PR2R of the second right magnetic member 12BR, which is the S-pole portion, as illustrated in the middle part of each of FIG. 8C and FIG. 8D.

(93) When the second lens holder 3B is moved rearward (in the X2 direction) from the reference position, the second magnetic sensor 6B is moved away from the second right protruding portion PR2R of the second right magnetic member 12BR, which is the S-pole portion, and partially faces the first left protruding portion PR1L of the second left magnetic member 12BL, which is the N-pole portion, as illustrated in the upper part of FIG. 8C and the left part of FIG. 8D.

(94) When the second lens holder 3B is moved forward (in the X1 direction) from the reference position, the second magnetic sensor 6B is moved away from the second right protruding portion PR2R of the second right magnetic member 12BR, which is the S-pole portion, and partially faces the second left protruding portion PR2L of the second left magnetic member 12BL, which is the N-pole portion, as illustrated in the lower part of FIG. 8C and the right part of FIG. 8D.

(95) By utilizing the above-described position detection mechanism DT, a controller (not illustrated), connected to the second magnetic sensor 6B via a conductive pattern formed on the substrate 4, can detect the position of the second lens holder 3B based on a voltage value output from the second magnetic sensor 6B. In the illustrated example, the controller is provided outside the housing HS, but may be provided within the housing HS.

(96) In the illustrated example, the magnetic field generating member MG is configured by the single bipolar permanent magnet. However, the magnetic field generating member MG may be configured by a combination of a plurality of bipolar permanent magnets, may be configured by one multipolar permanent magnet, or may be configured by a combination of a plurality of multipolar permanent magnets.

(97) Next, the movement of the lens holder 3 will be described with reference to FIG. 9. FIG. 9 is a perspective view of the lens holder 3, the shaft member 5, the receiving member 7, the lens body LS, and the magnetic field generating member MG. Specifically, the upper part of FIG. 9 illustrates the positional relationship between the members (the lens holder 3, the shaft member 5, the receiving member 7, the lens body LS, and the magnetic field generating member MG) when the first lens holder 3A is moved to the frontmost side (the X1 side) and the second lens holder 3B is moved to the rearmost side (the X2 side). The middle part of FIG. 9 illustrates the positional relationship between the members when the first lens holder 3A is moved to the frontmost side (the X1 side) and the second lens holder 3B is also moved to the frontmost side (the X1 side). The lower part of FIG. 9 illustrates the positional relationship between the members when the first lens holder 3A is moved to the rearmost side (the X2 side) and the second lens holder 3B is also moved to the rearmost side (the X2 side).

(98) As illustrated in FIG. 9, the lens holder driving apparatus 101 can separately move the first lens holder 3A and the second lens holder 3B along the optical axis direction.

(99) Next, the lens holder 3 will be described in detail with reference to FIG. 10. FIG. 10 is a bottom perspective view of the lens holder 3. The description with reference to FIG. 10 relates to the second lens holder 3B, but also applies analogously to the first lens holder 3A.

(100) As described above, the second lens holder 3B includes the second holding portion 31B, the second left shaft-receiving portion 32BL, and the second right shaft-receiving portion 32BR.

(101) The second holding portion 31B has a circular through hole RH through which the optical

axis OA passes. The circular through hole RH includes a front circular through hole RHF and a rear circular through hole RHB. In the illustrated example, the front circular through hole RHF is configured by a combination of a U-shaped groove DCF that opens upward (in the Z1 direction) and a U-shaped groove UCF that opens downward (in the Z2 direction). Specifically, the U-shaped groove DCF and the U-shaped groove UCF are disposed adjacent to each other in the front-rear direction (X-axis direction) and disposed facing each other in the upper-lower direction. Similarly, the rear circular through hole RHB is configured by a combination of a U-shaped groove DCB that opens upward (in the Z1 direction) and a U-shaped groove UCB that opens downward (in the Z2 direction). Specifically, the U-shaped groove DCB and the U-shaped groove UCB are disposed adjacent to each other in the front-rear direction (X-axis direction) and disposed facing each other in the upper-lower direction (Z-axis direction).

(102) Further, the second right shaft-receiving portion 32BR has a circular through hole TH through which the second shaft member 5B passes. The circular through hole TH includes a front circular through hole THF and a rear circular through hole THB. In the illustrated example, the front circular through hole THF is configured by a combination of a U-shaped groove DRF that opens upward (in the Z1 direction) and a U-shaped groove URF that opens downward (in the Z2 direction). Specifically, the U-shaped groove DRF and the U-shaped groove URF are disposed adjacent to each other in the front-rear direction (X-axis direction) and disposed facing each other in the upper-lower direction. Similarly, the rear circular through hole THB is configured by a combination of a U-shaped groove DRB that opens upward (in the Z1 direction) and a U-shaped groove URB that opens downward (in the Z2 direction). Specifically, the U-shaped groove DRB and the U-shaped groove URB are disposed adjacent to each other in the front-rear direction (X-axis direction) and disposed facing each other in the upper-lower direction (Z-axis direction).

(103) Further, the second left shaft-receiving portion 32BL has the notch CT through which the first shaft member 5A passes. In the illustrated example, the notch CT is configured by a combination of an upper L-shaped groove UL that opens downward (in the Z2 direction) and to the left (in the Y1 direction) and a lower L-shaped groove DL that opens upward (in the Z1 direction) and to the left (in the Y1 direction). Specifically, the upper L-shaped groove UL and the lower L-shaped groove DL are disposed adjacent to each other in the front-rear direction (X-axis direction) and disposed facing each other in the upper-lower direction (Z-axis direction).

(104) As described, the circular through hole TH is configured by the pair of U-shaped grooves, and the notch CT is configured by the pair of L-shaped grooves. Therefore, the second lens holder 3B can be manufactured by injection molding using a pair of molds separated in the upper-lower direction and not including a slide core. With this configuration, the manufacturing cost can be reduced.

(105) As described above, as illustrated in FIG. 3, the lens holder driving apparatus 101 according to an embodiment of the present invention includes the fixed member FB (base member 2), the lens holder 3 configured to hold the lens body LS, the shaft member 5 that is attached to the fixed member FB (base member 2) and extends along the optical axis direction so as to guide the lens holder 3, and the piezoelectric driver PD that includes the piezoelectric element 8 and is configured to move the lens holder 3 along the optical axis direction by a movement (circular motion) of the piezoelectric element 8. The piezoelectric element 8 extends in a first direction (Z-axis direction) intersecting the optical axis direction. The piezoelectric driver PD is provided on the fixed member FB (base member 2). The receiving member 7 is provided on the lens holder 3, is disposed facing the piezoelectric driver so as to contact the piezoelectric driver, extends in a second direction (X-axis direction) intersecting the first direction in which the piezoelectric element 8 extends, and is configured to receive the movement of the piezoelectric element 8. The piezoelectric driver PD includes the contact member 9 on a surface on the lens holder side of the piezoelectric element 8, and is configured to be preloaded toward the receiving member 7 by the preload member 13 such that the receiving member 7 contacts the contact member 9. The preload member 13 is provided on

the fixed member FB (base member 2). In the above-described embodiment, the lens holder driving apparatus **101** is configured to separately move two lens holders **3** (the first lens holder **3A** and the second lens holder **3B**) along the optical axis direction. However, the lens holder driving apparatus **101** may include one lens holder **3**, and may be configured to move the one lens holder **3** along the optical axis direction. Alternatively, the lens holder driving apparatus **101** may be configured to separately move three or more lens holders **3** along the optical axis direction.

(106) With the above configuration, ease of assembly of the lens holder driving apparatus **101** can be facilitated. That is, with the above-described configuration, the lens holder driving apparatus **101** can be more easily assembled. This is because the piezoelectric driver PD is provided on the fixed member FB (base member 2), and power can be supplied to the piezoelectric driver PD without using a member such as a flexible substrate that deforms in accordance with the movement of the lens holder **3**. In addition, one effect of the above-described configuration is that the movement of the lens holder **3** is not restricted by a member such as a flexible substrate that deforms in accordance with the movement of the lens holder **3**.

(107) Further, in the above-described embodiment, as illustrated in FIG. 4A and the FIG. 4B, the lens holder **3** may include the holding portion **31** configured to hold the lens body LS, and include the shaft-receiving portion **32** configured to receive the shaft member **5**. In this case, the receiving member **7** may be composed of a different material from the lens holder **3**. In the example illustrated in FIG. 4A and FIG. 4B, the receiving member **7** is formed of a metal, and the lens holder **3** is formed of a synthetic resin. The receiving member **7** is fixed to the shaft-receiving portion **32**. The receiving member **7** is longer than the holding portion **31** in the optical axis direction. In addition, the shaft-receiving portion **32** to which the receiving member **7** is fixed is longer than the holding portion **31** in the optical axis direction. Further, the width of the receiving member **7** in the optical axis direction is greater than the width of the movable range of the lens holder **3** in the optical axis direction. In the example illustrated in FIG. 4A, the first lens holder **3A** includes the first holding portion **31A** configured to hold the first lens body LS1, the first left shaft-receiving portion **32AL** configured to receive the first shaft member **5A**, and the first right shaft-receiving portion **32AR** configured to receive the second shaft member **5B**. Further, the second lens holder **3B** includes the second holding portion **31B** configured to hold the second lens body LS2, the second left shaft-receiving portion **32BL** configured to receive the first shaft member **5A**, and the second right shaft-receiving portion **32BR** configured to receive the second shaft member **5B**. The first receiving member **7A** is longer than the first holding portion **31A** in the optical axis direction. In addition, the first left shaft-receiving portion **32AL** to which the first receiving member **7A** is fixed is longer than the first holding portion **31A** in the optical axis direction. Similarly, the second receiving member **7B** is longer than the second holding portion **31B** in the optical axis direction. In addition, the second right shaft-receiving portion **32BR** to which the second receiving member **7B** is fixed is longer than the second holding portion **31B** in the optical axis direction. Further, the width W3 of the first receiving member **7A** in the optical axis direction is greater than the width of the movable range of the first lens holder **3A** in the optical axis direction. The width W7 of the second receiving member **7B** in the optical axis direction is greater than the width of the movable range of the second lens holder **3B** in the optical axis direction.

(108) One effect of the above-described configuration is that the width of the movable range of the first lens holder **3A** in the optical axis direction can be made greater than at least the width of the first holding portion **31A** in the optical axis.

(109) In the above-described embodiment, the shaft member **5** includes the first shaft member **5A** and the second shaft member **5B**. As illustrated in FIG. 7, the receiving member **7** is provided at a position away from a virtual plane VP. Specifically, the first receiving member **7A** is provided at a position where its axis **7AX** is not located on the virtual plane VP, and the second receiving member **7B** is provided at a position where its axis **7BX** is not located on the virtual plane VP. In the present embodiment, the axis **7AX** and the axis **7BX** are configured to be parallel to the virtual

plane VP. The virtual plane VP is a virtual plane that includes the axis (axis 5AX) of the first shaft member 5A and the axis (axis 5BX) of the second shaft member 5B, which are parallel to each other. In the example illustrated in FIG. 7, a first movable body, including the first lens holder 3A, the first receiving member 7A, and the first magnetic field generating member MG1, is configured such that its center of gravity is located on the virtual plane VP. With this configuration, it is possible to suppress the generation of a torque around the center of gravity when the first movable body is moved along the shaft member 5. The same applies to a second movable body including the second lens holder 3B, the second receiving member 7B, and the second magnetic field generating member MG2. Specifically, as illustrated in FIG. 3, the base member 2, serving as the fixed side member FB, includes the bottom surface portion 2B. Further, as illustrated in FIG. 4B, the lens holder 3 has an opening at the top thereof, and the lens body LS can be inserted (accommodated) into the lens holder 3 through the opening. As illustrated in FIG. 7, the receiving member 7 is provided at a position offset upward (to the Z1 side) from the virtual plane VP. Note that the receiving member 7 may be provided at a position offset downward (to the Z2 side) from the virtual plane VP.

(110) One effect of the above-described configuration is that backlash of the lens holder 3 can be suppressed. In the example illustrated in FIG. 7, the preload force (preload force F1 indicated by a dotted arrow) of the second preload member 13B that preloads the second receiving member 7B, attached to the second lens holder 3B, toward the Y1 side generates a torque (preload torque TQ indicated by a dash-dot arrow) that rotates the second lens holder 3B around the axis (axis 5BX) of the second shaft member 5B. The preload torque TQ acts so as to press the second left shaft-receiving portion 32BL of the second lens holder 3B against the first shaft member 5A from above. In addition to the preload torque TQ, a torque (a self-weight torque) acts on the second lens holder 3B to rotate the second lens holder 3B around the (axis 5BX) of the second shaft member 5B due to the self-weight of the second lens holder 3B. The second preload member 13B is configured such that the magnitude of the preload torque TQ generated by the preload force F1 is larger than the magnitude of the self-weight torque. Therefore, a combined torque obtained by combining the preload torque TQ and the self-weight torque always acts so as to press the second left shaft-receiving portion 32BL of the second lens holder 3B against the first shaft member 5A, regardless of the orientation of the lens holder driving apparatus 101 (even if the lens holder driving apparatus 101 is turned upside down). That is, not only when the preload torque TQ and the self-weight torque are in the same direction, but also when the preload torque TQ and the self-weight torque are in opposite directions, the combined torque always acts so as to press the second left shaft-receiving portion 32BL of the second lens holder 3B against the first shaft member 5A. As a result, the second left shaft-receiving portion 32BL of the second lens holder 3B always contacts the first shaft member 5A regardless of the orientation of the lens holder driving apparatus 101 (even if the lens holder driving apparatus is turned upside down), thereby suppressing backlash between the second left shaft-receiving portion 32BL and the first shaft member 5A. The same applies to backlash between the second right shaft-receiving portion 32BR of the second lens holder 3B and the second shaft member 5B, backlash between the first right shaft-receiving portion 32AR of the first lens holder 3A and the second shaft member 5B, and backlash between the first left shaft-receiving portion 32AL of the first lens holder 3A and the first shaft member 5A.

(111) Further, one effect of the above-described configuration is that the lens body LS can be easily attached to the lens holder 3 can be obtained. Typically, the lens body LS is attached to the lens holder 3 in a state in which the lower member LM (excluding the lens body LS) is placed on a work table in an orientation as illustrated in FIG. 1B. In this state, the first left shaft-receiving portion 32AL of the first lens holder 3A is pressed against the first shaft member 5A, and the first right shaft-receiving portion 32AR of the first lens holder 3A is pressed against the second shaft member 5B from above. Further, the second right shaft-receiving portion 32BR of the second lens holder 3B is pressed against the second shaft member 5B, and the second left shaft-receiving

portion **32BL** of the second lens holder **3B** is pressed against the first shaft member **5A** from above. That is, the position of the lens holder **3** does not change before and after the lens body **LS** is attached to the lens holder **3**.

(112) Further, in the above-described embodiment, each of the receiving member **7** and the contact member **9** is formed of a metal. The receiving member **7** is harder than the contact member **9**.

(113) One effect of the above-described configuration is that the receiving member **7** and the contact member **9** are not readily worn, as compared to when at least one of the receiving member **7** and the contact member **9** are formed of a synthetic resin, for example. Further, with the above-described configuration, the contact member **9** is configured to be softer than (readily curved as compared to) the receiving member **7**. Therefore, one effect of the above-described configuration is that the flexural vibration (circular motion) of the piezoelectric element **8** can be prevented from being excessively reduced.

(114) Further, in the above-described embodiment, the preload member **13** is a leaf spring member extending in the optical axis direction. Specifically, as illustrated in FIG. **5B**, the preload member **13** includes a pair of fixed portions **13F**, a support portion **13S**, and a pair of elastically deformable portions **13E**. The fixed portions **13F** are fixed to the base member **2**, which serves as the fixed member **FB**, the support portion **13S** supports the piezoelectric driver **PD**, and each of the elastically deformable portions **13E** is elastically deformable and provided between a corresponding fixed portion **13F** and the support portion **13S**.

(115) One effect of the above-described configuration is that the dimensional tolerance of the preload member **13** can be reduced as compared to a configuration that utilizes a leaf spring member extending in a direction (Z-axis direction) perpendicular to the optical axis direction. This is because the housing **HS** is configured such that the dimension of the housing **HS** in the optical axis direction (X-axis direction) is larger than the dimension (height) of the housing **HS** in the Z-axis direction. That is, the dimension of an elastically deformable portion **13E** (the distance between a fixed portion **13F** and a support portion **13S**) in the optical axis direction can be increased as compared to a configuration that utilizes a leaf spring member extending in a direction (Z-axis direction) perpendicular to the optical axis direction. As the dimension of the elastically deformable portion **13E** in the optical axis direction increases, the influence of deviation from the reference dimension of the elastically deformable portion **13E** on the spring constant of the elastically deformable portion **13E** decreases.

(116) Further, in the above-described embodiment, as illustrated in FIG. **10**, the lens holder **3** has a plurality of through holes. The plurality of through holes include the circular through hole **RH** through which the optical axis **OA** passes and the circular through hole **TH** through which the shaft member **5** passes. The circular through hole **RH** includes the front circular through hole **RHF** and the rear circular through hole **RHB**. The circular through hole **TH** includes the front circular through hole **THF** and the rear circular through hole **THB**. Each of the through holes is configured by a combination of a first groove that opens upward and a second groove that opens downward. Specifically, the front circular through hole **RHF** is configured by a combination of the U-shaped groove **DCF** that opens upward (in the **Z1** direction) and the U-shaped groove **UCF** that opens downward (in the **Z2** direction). The rear circular through hole **RHB** is configured by a combination of the U-shaped groove **DCB** that opens upward (in the **Z1** direction) and the U-shaped groove **UCB** that opens downward (in the **Z2** direction). The front circular through hole **THF** is configured by a combination of the U-shaped groove **DRF** that opens upward (in the **Z1** direction) and the U-shaped groove **URF** that opens downward (in the **Z2** direction). The rear circular through hole **THB** is configured by a combination of the U-shaped groove **DRB** that opens upward (in the **Z1** direction) and the U-shaped groove **URB** that opens downward (in the **Z2** direction). That is, in the above-described embodiment, the lens holder **3** is manufactured by injection molding using a pair of molds separated in the upper-lower direction (Z-axis direction) and not including a slide core.

(117) Accordingly, one effect of the above-described configuration is that the manufacturing cost can be reduced as compared to when the lens holder **3** is manufactured by injection molding using a mold that includes a slide core. This is because, for example, a decrease in manufacturing yield or a decrease in manufacturing efficiency due to mold-related defects can be suppressed.

(118) Further, in the above-described embodiment, as illustrated in FIG. **3**, the magnetic field generating member MG is provided on the lens holder **3** and extends in the optical axis direction. As illustrated in FIG. **8A**, the magnetic field generating member MG includes a magnet **11** (second magnet **11B**), a first magnetic member (second left magnetic member **12BL**), and a second magnetic member (second right magnetic member **12BR**). The magnet **11** is magnetized such that first end portion (an end portion on the Y1 side (a left end portion)), in a third direction (Y-axis direction) perpendicular to the optical axis direction, of the magnet **11** becomes a S-pole and a second end portion (an end portion on the Y2 side (a right end portion)) of the magnet **11** becomes an N-pole. The first magnetic member (second left magnetic member **12BL**) is fixed to the first end portion of the magnet, and the second magnetic member (second right magnetic member **12BR**) is fixed to the second end portion of the magnet. The first magnetic member (second left magnetic member **12BL**) has a plurality of first protruding portions (first to third left protruding portions PR1L to PR3L) that protrude toward the second end portion of the magnet **11**. The second magnetic member (second right magnetic member **12BR**) has a plurality of second protruding portions (first to third right protruding portions PR1R to PR3R) that protrude toward the first end portion of the magnet **11**. The first protruding portions (first to third left protruding portions PR1L to PR3L) and the second protruding portions (first to third right protruding portions PR1R to PR3R) protrude in opposite directions, and are alternately arranged in the optical axis direction (X-axis direction) in which the magnetic field generating member MG extends. In the illustrated example, the first magnetic member (second left magnetic member **12BL**) has the plurality of first protruding portion (first to third left protruding portions PR1L to PR3L) that protrude in the first direction (Z-axis direction) perpendicular to the optical axis direction and perpendicular to the third direction (Y-axis direction). The second magnetic member (second right magnetic member **12BR**) has the plurality of second protruding portion (first to third right protruding portions PR1R to PR3R) that protrude in the first direction (Z-axis direction). The first protruding portions (first to third left protruding portions PR1L to PR3L) and the second protruding portions (first to third right protruding portions PR1R to PR3R) protrude in the opposite directions, and are alternately arranged in the optical axis direction (X-axis direction) in which the magnetic field generating member MG extends. The magnetic sensor **6** is provided on the base member **2** (substrate **4**), which serves as the fixed member FB, and is configured to detect a magnetic field generated by each of the first protruding portion (first to third left protruding portions PR1L to PR3L) and of the second protruding portions (first to third right protruding portions PR1R to PR3R).

(119) This configuration has the effect that the manufacturing cost of the lens holder driving apparatus **101** can be reduced. This is because the movement of the lens holder **3** in the optical axis direction can be detected by using the magnetic field generating member MG configured by a combination of the magnetic member **12** and the magnet **11** magnetized to two poles, without using a permanent magnet magnetized to multiple poles.

(120) Further, in the above-described embodiment, as illustrated in FIG. **1B**, the lens body LS includes the first lens body LS1 and the second lens body LS2. The first lens body LS1 and the second lens body LS2 have the common optical axis OA. As illustrated in FIG. **3**, the lens holder **3** includes the first lens holder **3A** and the second lens holder **3B**. The first lens holder **3A** is configured to hold the first lens body LS1, and the second lens holder is configured to hold the second lens body LS2. The shaft member **5** includes the first shaft member **5A** and the second shaft member **5B**. The piezoelectric driver PD includes the first piezoelectric driver PD1 and the second piezoelectric driver PD2. The first piezoelectric driver PD1 includes the first piezoelectric element **8A** and is configured to move the first lens holder **3A** along the optical axis direction by a

movement (circular motion) of the first piezoelectric element **8A**, and the second piezoelectric driver **PD2** includes the second piezoelectric element **8B** and is configured to move the second lens holder **3B** along the optical axis direction by a movement (circular motion) of the second piezoelectric element **8B**. The first piezoelectric element **8A** and the second piezoelectric element **8B** extend in the first direction (Z-axis direction) intersecting the optical axis direction. The first lens holder **3A** and the second lens holder **3B** are disposed facing each other in the optical axis direction. The first piezoelectric driver **PD1** and the second piezoelectric driver **PD2** are both provided on the base member **2**, which serves as the fixed member **FB**. The first piezoelectric driver **PD1** and the second piezoelectric driver **PD2** are apart from each other, with the first shaft member **5A** and the second shaft member **5B** interposed therebetween. Specifically, as illustrated in FIG. 3, the first receiving member **7A** is provided on the first lens holder **3A**, is disposed facing the first piezoelectric driver **PD1** so as to contact the first piezoelectric driver **PD1**, extends in the X-axis direction intersecting the first direction (Z-axis direction) in which the first piezoelectric element **8A** extends, and is configured to receive the movement of the first piezoelectric element **8A**. As illustrated in FIG. 4A, the first lens holder **3A** includes a first portion (first left shaft-receiving portion **32AL**) located near the first shaft member **5A** and a second portion (first right shaft-receiving portion **32AR**) located near the second shaft member **5B**. The first receiving member **7A** is provided on the first portion (first left shaft-receiving portion **32AL**), located near the first shaft member **5A**, of the first lens holder **3A**. Similarly, as illustrated in FIG. 3, the second receiving member **7B** is provided on the second lens holder **3B**, is disposed facing the second piezoelectric driver **PD2** so as to contact the second piezoelectric driver **PD2**, extends in the X-axis direction intersecting the first direction (Z-axis direction) in which the second piezoelectric element **8B** extends, and is configured to receive the movement of the second piezoelectric element **8B**. As illustrated in FIG. 4A, the second lens holder **3B** includes a third portion (second left shaft-receiving portion **32BL**) located near the first shaft member **5A** and a fourth portion (second right shaft-receiving portion **32BR**) located near the second shaft member **5B**. The second receiving member **7B** is provided on the fourth portion (second right shaft-receiving portion **32BR**), located near the second shaft member **5B**, of the second lens holder **3B**. The first piezoelectric driver **PD1** includes the first contact member **9A** on a surface on the first lens holder **3A** side (Y2 side) of the first piezoelectric element **8A**, and is configured to be preloaded toward the first receiving member **7A** (Y2 side) by the first preload member **13A** such that the first receiving member **7A** contacts the first contact member **9A**. The first preload member **13A** is provided on the base member **2**, which serves as the fixed member **FB**. Similarly, the second piezoelectric driver **PD2** includes the second contact member **9B** on a surface on the second lens holder **3B** side (Y1 side) of the second piezoelectric element **8B**, and is configured to be preloaded toward the second receiving member **7B** (Y1 side) by the second preload member **13B** such that the second receiving member **7B** contacts the second contact member **9B**. The second preload member **13B** is provided on the base member **2**, which serves as the fixed member **FB**.

(121) One effect of the above-described configuration is that ease of assembly of the lens holder driving apparatus **101** can be facilitated. That is, with the above-described configuration, the lens holder driving apparatus **101** can be more easily assembled. This is because the first piezoelectric driver **PD1** is provided on the fixed member **FB** (base member **2**), and power can be supplied to the first piezoelectric driver **PD1** without using a member such as a flexible substrate that deforms in accordance with the movement of the first lens holder **3A**. Similarly, this is because the second piezoelectric driver **PD2** is provided on the fixed member **FB** (base member **2**), and power can be supplied to the second piezoelectric driver **PD2** without using a member such as a flexible substrate that deforms in accordance with the movement of the first lens holder **3A**. Further, one effect of the above-described configuration is that the movement of the first lens holder **3A** is not restricted by a member such as a flexible substrate that deforms in accordance with the movement of the first lens holder **3A**. In addition, one effect of the above-described configuration is that the movement of the

second lens holder **3B** is not restricted by a member such as a flexible substrate that deforms in accordance with the movement of the second lens holder **3B**.

(122) Further, with the above-described configuration, the first piezoelectric driver **PD1** and the second piezoelectric driver **PD2** are provided on the fixed member **FB** (base member **2**) at positions away from each other. Accordingly, easy of assembly can be facilitated.

(123) In the above-described embodiment, as illustrated in FIG. **4A**, the first lens holder **3A** includes the first holding portion **31A** that holds the first lens body **LS1**. The first receiving member **7A** is longer than the first holding portion **31A** in the optical axis direction. Similarly, as illustrated in FIG. **4A**, the second lens holder **3B** includes the second holding portion **31B** configured to hold the second lens body **LS2**. The second receiving member **7B** is longer than the second holding portion **31B** in the optical axis direction. At least a part of the first receiving member **7A** and at least a part of the second receiving member **7B** overlap in the third direction (Y-axis direction) perpendicular to the optical axis direction in a state in which the first holding portion **31A** and the second holding portion **31B** are in proximity to each other. In the state illustrated in FIG. **4A**, the first receiving member **7A** and the second receiving member **7B** overlap over the width **W9** in the third direction (Y-axis direction) perpendicular to the optical axis direction.

(124) One effect of the above-described configuration is that the amount of movement of each of the first lens holder **3A** and the second lens holder **3B** in the optical axis direction can be increased within the limited length of the shaft member **5** in the optical axis direction. While the width **W1** of the first left shaft-receiving portion **32AL** is set to be greater than the width **W2** of the first holding portion **31A**, and the width **W5** of the second right shaft-receiving portion **32BR** is set to be greater than the width **W6** of the second holding portion **31B**, the first lens body **LS1** and the second lens body **LS2** can approach each other to the extent that the first lens body **LS1** and the second lens body **LS2** are nearly in contact with each other.

(125) As illustrated in FIG. **7**, each of the first receiving member **7A** and the second receiving member **7B** may be provided at a position away from the virtual plane **VP** that includes the axis of the first shaft member **5A** and the axis of the second shaft member **5B**.

(126) One effect of the above-described configuration is that backlash of each of the first lens holder **3A** and the second lens holder **3B** can be suppressed. This is because the lens holder driving apparatus **101** can have a configuration in which a torque acting on the first lens holder **3A** always acts so as to press the first lens holder **3A** against the shaft member **5**, and a torque act on the second lens holder **3B** always acts so as to press the second lens holder **3B** against the shaft member **5**, regardless of the orientation of the lens holder driving apparatus **101** (even if the lens holder driving apparatus **101** is turned upside down).

(127) In the above-described embodiment, as illustrated in FIG. **3**, the base member **2**, serving as the fixed member **FB**, includes the bottom surface portion **2B**. Further, the first lens holder **3A** has an opening at the top thereof, and the first lens body **LS1** can be inserted into the first lens holder **3A** through the opening. As illustrated in FIG. **7**, the first receiving member **7A** is located above (on the **Z1** side relative to) the virtual plane **VP**. Similarly, as illustrated in FIG. **4B**, the second lens holder **3B** has an opening at the top thereof, and the second lens body **LS2** can be inserted into the second lens holder **3B** through the opening. As illustrated in FIG. **7**, the second receiving member **7B** is located above (on the **Z1** side relative to) the virtual plane **VP**.

(128) One effect of the above-described configuration is that the first lens body **LS1** can be easily attached to the first lens holder **3A**. This is because the first left shaft-receiving portion **32AL** of the first lens holder **3A** is pressed against the first shaft member **5A**, and the first right shaft-receiving portion **32AR** of the first lens holder **3A** is pressed against the second shaft member **5B** from above. That is, the lens holder driving apparatus **101** is configured to prevent backlash between the first lens holder **3A** and each of the first shaft member **5A** and the second shaft member **5B** before and after the first lens body **LS1** is attached to the first lens holder **3A**. The same applies to a case where the second lens body **LS2** is attached to the second lens holder **3B**.

(129) In the above-described embodiment, the first preload member **13A** is a leaf spring member. As illustrated in FIG. 5B, the first preload member **13A** includes a pair of first fixed portions (fixed portions **13F**), a first support portion (support portion **13S**), and a pair of first elastically deformable portions (elastically deformable portions **13E**). The first fixed portions (fixed portions **13F**) are fixed to the base member **2**, which serves as the fixed member FB, the first support portion (support portion **13S**) supports the first piezoelectric driver PD1, and each of the first elastically deformable portions (elastically deformable portions **13E**) is elastically deformable and provided between a corresponding first fixed portion (fixed portion **13F**) and the first support portion (support portion **13S**). Similarly, the second preload member **13B** is a leaf spring member. The second preload member **13B** includes a pair of second fixed portions (fixed portions **13F**), a second support portion (support portion **13S**), and a pair of second elastically deformable portions (elastically deformable portions **13E**). The second fixed portions (fixed portions **13F**) are fixed to the base member **2**, which serves as the fixed member FB, the second support portion (support portion **13S**) supports the second piezoelectric driver PD2, and each of the second elastically deformable portions (elastically deformable portions **13E**) is elastically deformable and provided between a corresponding second fixed portion (fixed portion **13F**) and the second support portion (support portion **13S**).

(130) One effect of the above-described configuration is that the dimensional tolerance of each of the first preload member **13A** and the second preload member **13B** can be reduced as compared to a configuration that utilizes a leaf spring member extending in a direction (Z-axis direction) perpendicular to the optical axis direction. This is because the housing HS is configured to have a larger dimension in the optical axis direction (X-axis direction) than a dimension (height) in the Z-axis direction. That is, with the above-described configuration, the dimension of an elastically deformable portion **13E** in the optical axis direction (the distance between a fixed portion **13F** and a support portion **13S**) can be increased as compared to a configuration that utilizes a leaf spring member extending in a direction (Z-axis direction) perpendicular to the optical axis direction. As the dimension of the elastically deformable portion **13E** in the optical axis direction increases, the influence of deviation from the reference dimension of the elastically deformable portion **13E** on the spring constant of the elastically deformable portion **13E** decreases.

(131) In the above-described embodiment, the first magnetic field generating member MG1 is provided on the first shaft member **5A** side of the first lens holder **3A**, and extends in the optical axis direction. The second magnetic field generating member MG2 is provided on the second shaft member **5B** side of the second lens holder **3B**, and extends in the optical axis direction. The first magnetic sensor **6A** is provided on the base member **2** (substrate **4**), which serves as the fixed member FB, and is configured to detect a magnetic field generated by the first magnetic field generating member MG1. The second magnetic sensor **6B** is provided on the base member **2** (substrate **4**), and is configured to detect a magnetic field generated by the second magnetic field generating member MG2.

(132) One effect of the above-described configuration is that magnetic mutual interference between the first position detection mechanism DT1, configured by the first magnetic field generating member MG1 and the first magnetic sensor **6A**, and the second position detection mechanism DT2, configured by the second magnetic field generating member MG2 and the second magnetic sensor **6B**, can be suppressed.

(133) According to an embodiment of the present invention, a lens holder driving apparatus that can be easily assembled is provided.

(134) The embodiment of the present invention has been described in detail above. However, the present invention is not limited to the particulars of the above-described embodiment. Various modifications and substitutions may be applied to the above-described embodiment without departing from the scope of the present invention. Furthermore, the separately described features may be suitably combined as long as no technical contradiction occurs.

(135) For example, in the above-described embodiment, the circular through hole TH is configured by a combination of two U-shaped grooves. However, one or both of the U-shaped grooves may be V-shaped grooves. The same applies to the circular through hole RH.

Claims

1. A lens holder driving apparatus comprising: a fixed member; a lens holder configured to hold a lens body; a shaft member that is provided on the fixed member and extends along an optical axis direction so as to guide the lens holder; a piezoelectric driver that includes a piezoelectric element and is configured to move the lens holder along the optical axis direction by a movement of the piezoelectric element, the piezoelectric element extending in a first direction intersecting the optical axis direction; a receiving member that is provided on the lens holder; and a preload member that is provided on the fixed member; wherein the piezoelectric driver is provided on the fixed member, wherein the receiving member is disposed facing the piezoelectric driver so as to contact the piezoelectric driver, extends in a second direction intersecting the first direction in which the piezoelectric element extends, and is configured to receive the movement of the piezoelectric element, wherein the piezoelectric element is configured to generate a circular motion about a rotation axis parallel to the direction in which the piezoelectric element extends, and the receiving member is moved forward and rearward in the optical axis direction by the circular motion, and wherein the piezoelectric driver includes a contact member on a surface on a lens holder side of the piezoelectric element, and is configured to be preloaded toward the receiving member by the preload member such that the receiving member contacts the contact member.
2. The lens holder driving apparatus according to claim 1, wherein the lens holder includes a holding portion configured to hold the lens body, and includes a shaft-receiving portion configured to receive the shaft member, the receiving member is composed of a different material from the lens holder, fixed to the shaft-receiving portion, and longer than the holding portion in the optical axis direction, and the shaft-receiving portion is longer than the holding portion in the optical axis direction.
3. The lens holder driving apparatus according to claim 1, wherein the shaft member includes a first shaft member and a second shaft member, and the receiving member is provided at a position away from a virtual plane that includes an axis of the first shaft member and an axis of the second shaft member.
4. The lens holder driving apparatus according to claim 3, wherein the fixed member includes a bottom surface portion, the lens holder has an opening at a top thereof, and the lens body is inserted into the lens holder through the opening, and the receiving member is located above the virtual plane.
5. The lens holder driving apparatus according to claim 1, wherein each of the receiving member and the contact member is formed of a metal, and the receiving member is harder than the contact member.
6. The lens holder driving apparatus according to claim 1, wherein the preload member is a leaf spring member extending in the optical axis direction, and includes a pair of fixed portions, a support portion, and a pair of elastically deformable portions, the pair of fixed portions being fixed to the fixed member, the support portion supporting the piezoelectric driver, and each of the elastically deformable portions being elastically deformable and provided between a corresponding fixed portion and the support portion.
7. The lens holder driving apparatus according to claim 1, wherein the lens holder has a plurality of through holes in the optical axis direction, and each of the through holes is configured by a combination of a first groove that opens upward and a second groove that opens downward.
8. The lens holder driving apparatus according to claim 1, further comprising a magnetic field generating member and a magnetic sensor, wherein the magnetic field generating member is

provided on the lens holder and extends in the optical axis direction, the magnetic field generating member includes a magnet, a first magnetic member, and a second magnetic member, the magnet being magnetized such that a first end portion, in a third direction perpendicular to the optical axis direction, of the magnet becomes a S-pole and a second end portion of the magnet becomes an N-pole, the first magnetic member being fixed to the first end portion of the magnet, and the second magnetic member being fixed to the second end portion of the magnet, the first magnetic member has a plurality of first protruding portions that protrude toward the second end portion of the magnet, the second magnetic member has a plurality of second protruding portions that protrude toward the first end portion of the magnet, the first protruding portions and the second protruding portions protrude in opposite directions, and are alternately arranged in the optical axis direction in which the magnetic field generating member extends, and the magnetic sensor is provided on the fixed member, and is configured to detect a magnetic field generated by each of the first protruding portions and the second protruding portions.

9. The lens holder driving apparatus according to claim 1, wherein the lens body includes a first lens body and a second lens body, the first lens body and the second lens body having a common optical axis, the lens holder includes a first lens holder and a second lens holder, the first lens holder being configured to hold the first lens body, and the second lens holder being configured to hold the second lens body, the shaft member includes a first shaft member and a second shaft member, the piezoelectric driver includes a first piezoelectric driver and a second piezoelectric driver, the first piezoelectric driver including a first piezoelectric element and being configured to move the first lens holder along the optical axis direction by a movement of the first piezoelectric element, and the second piezoelectric driver including a second piezoelectric element and being configured to move the second lens holder along the optical axis direction by a movement of the second piezoelectric element, the first piezoelectric element and the second piezoelectric element extending in the first direction intersecting the optical axis direction, the first lens holder and the second lens holder are disposed facing each other in the optical axis direction, the first piezoelectric driver and the second piezoelectric driver are both provided on the fixed member, the receiving member includes a first receiving member and a second receiving member, the first receiving member being provided on the first lens holder and the second receiving member being provided on the second lens holder, the first receiving member is disposed facing the first piezoelectric driver so as to contact the first piezoelectric driver, extends in the second direction intersecting the first direction in which the first piezoelectric element extends, and is configured to receive the movement of the first piezoelectric element, the first lens holder includes a first portion located near the first shaft member and a second portion located near the second shaft member, the first receiving member is provided on the first portion, located near the first shaft member, of the first lens holder, the second receiving member is disposed facing the second piezoelectric driver so as to contact the second piezoelectric driver, extends in the second direction intersecting the first direction in which the second piezoelectric element extends, and is configured to receive the movement of the second piezoelectric element, the second lens holder includes a third portion located near the first shaft member and a fourth portion located near the second shaft member, the second receiving member is provided on the fourth portion, located near the second shaft member, of the second lens holder, the preload member includes a first preload member and a second preload member, the first preload member and the second preload member being provided on the fixed member, the first piezoelectric driver includes a first contact member on a surface, on a first lens holder side of the first piezoelectric element, and is configured to be preloaded toward the first receiving member by the first preload member, such that the first receiving member contacts the first contact member, and the second piezoelectric driver includes a second contact member on a surface, on a second lens holder side of the second piezoelectric element, and is configured to be preloaded toward the second receiving member by the second preload member, such that the second receiving member contacts the second contact member.

10. The lens holder driving apparatus according to claim 9, wherein the first lens holder includes a first holding portion configured to hold the first lens body, the first receiving member is longer than the first holding portion in the optical axis direction, the second lens holder includes a second holding portion configured to hold the second lens body, the second receiving member is longer than the second holding portion in the optical axis direction, and at least a part of the first receiving member and at least a part of the second receiving member overlap in a third direction perpendicular to the optical axis direction in a state in which the first holding portion and the second holding portion are in proximity to each other.

11. The lens holder driving apparatus according to claim 9, wherein each of the first receiving member and the second receiving member is provided at a position away from a virtual plane that includes an axis of the first shaft member and an axis of the second shaft member.

12. The lens holder driving apparatus according to claim 11, wherein the fixed member includes a bottom surface portion, the first lens holder has a first opening at a top thereof, and the first lens body is inserted into the first lens holder through the first opening, the first receiving member is located above the virtual plane, the second lens holder has a second opening at a top thereof, and the second lens body is inserted into the second lens holder through the second opening, and the second receiving member is located above the virtual plane.

13. The lens holder driving apparatus according to claim 9, wherein the first preload member is a leaf spring member extending in the optical axis direction, and includes a pair of first fixed portions, a first support portion, and a pair of first elastically deformable portions, the first fixed portions being fixed to the fixed member, the first support portion supporting the first piezoelectric driver, and each of the first elastically deformable portions being elastically deformable and provided between a corresponding first fixed portion and the first support portion, and wherein the second preload member is a leaf spring member extending in the optical axis direction, and includes a pair of second fixed portions, a second support portion, and a pair of second elastically deformable portions, the second fixed portions being fixed to the fixed member, the second support portion supporting the second piezoelectric driver, and each of the second elastically deformable portions being elastically deformable and provided between a corresponding second fixed portion and the second support portion.

14. The lens holder driving apparatus according to claim 9, further comprising a first magnetic field generating member, a second magnetic field generating member, a first magnetic sensor, and a second magnetic sensor, wherein the first magnetic field generating member is provided on a first shaft member side of the first lens holder and extends in the optical axis direction, the second magnetic field generating member is provided on a second shaft member side of the second lens holder and extends in the optical axis direction, the first magnetic sensor is provided on the fixed member and is configured to detect a magnetic field generated by the first magnetic field generating member, and the second magnetic sensor is provided on the fixed member and is configured to detect a magnetic field generated by the second magnetic field generating member.

15. The lens holder driving apparatus according to claim 1, wherein the piezoelectric element has a two-layer structure including a first layer configured to generate a first flexural vibration in a first plane and a second layer configured to generate a second flexural vibration in a second plane, said first plane being perpendicular to the second plane, and the piezoelectric element is configured to generate the circular motion by combining the first flexural vibration and the second flexural vibration.
